

Corporate Learning Progress, Intervention & Compliance Tracking System

Solution Proposal & Case Study Document

Version: 1.0 | Date: 22-May-2026

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1. Executive Summary

Organisations are accountable for employee learning outcomes, training attendance and compliance with regulatory bodies. However, learning signals are typically fragmented across LMS platforms, assessment systems and trainer notes, leading to late detection of gaps. This solution consolidates learning data into a unified employee learning profile, evaluates configurable competency and attendance rules to identify at-risk learners early, tracks interventions (remedial sessions, coaching/mentoring) and provides audit-ready compliance reporting.

1.1 Goals

- Aggregate training attendance, assessment scores and competency milestones into a single learning profile per employee.
- Provide configurable rule-based risk detection at employee and competency levels with explainable reasons.
- Enable intervention assignment and outcome tracking to measure effectiveness.
- Deliver compliance-ready reports with audit trails (rule versions, user actions, timestamps).

2. Problem Statement

Learning and compliance accountability requires timely visibility into participation, knowledge acquisition and competency progression. In many organisations, evidence is spread across multiple tools, causing delayed identification of employees falling behind, and interventions are reactive and inconsistently tracked.

2.1 Key Challenges

- Fragmented data across systems results in incomplete or inconsistent employee learning views.
- Competency progression is not just course completion; it includes levels, milestones, due dates and evidence.
- Rules are often embedded in spreadsheets or people's knowledge, making them hard to audit and change.
- Interventions are not tracked with outcomes, making it difficult to demonstrate effectiveness to auditors.

3. Scope and Non-Scope

3.1 In Scope

- Ingestion of training attendance records via API.
- Ingestion of periodic assessment scores via API.
- Ingestion of competency-level milestones via API.
- Rule-based risk identification using configurable rule definitions.

- Intervention tracking for remedial sessions and coaching/mentoring.
- Dashboards and compliance-ready reporting with audit evidence.

3.2 Out of Scope (Not a full LMS)

- Course catalogue management, content authoring/hosting, enrolment workflows.
- Full learning delivery experience (video streaming, SCORM player, etc.).
- Payroll, HRIS master data management (assumed via existing HR system).

4. Users and Personas

L&D Administrator

Defines mandatory learning programmes, configures risk rules, monitors compliance and produces audit reports.

Trainer / Coach / Mentor

Reviews at-risk learners, assigns interventions, records coaching outcomes and follow-up evidence.

Compliance Officer

Needs verifiable evidence of training completion, rule versions used, and audit trails.

Employee

Views personal learning progress, required milestones and assigned interventions.

5. Solution Overview

The system operates as a consolidation and risk/intervention layer on top of existing learning sources.

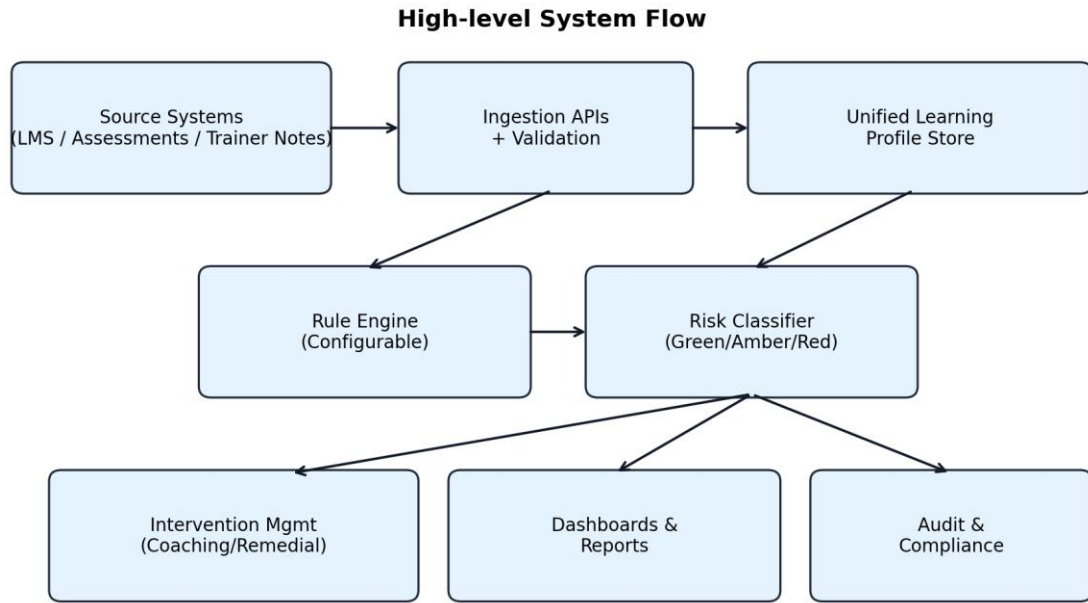


Figure 1: High-level system flow

6. Functional Requirements

6.1 Ingestion APIs

Three ingestion endpoints provide a consistent, auditable way to ingest learning signals.

6.1.1 Attendance Ingestion (API)

Endpoint: POST /ingest/attendance

- Key fields:
 - employeeId
 - trainingId
 - trainingType (MANDATORY/ELECTIVE)
 - sessionDate
 - status (ATTENDED/ABSENT)
 - durationMinutes
 - trainerId
 - sourceSystem
 - eventId (idempotency key)

6.1.2 Assessment Ingestion (API)

Endpoint: POST /ingest/assessments

- Key fields:
 - employeeId
 - assessmentId
 - competencyId
 - score
 - maxScore
 - attempt
 - assessmentDate
 - proctor/trainerId (optional)
 - sourceSystem
 - eventId

6.1.3 Milestone Ingestion (API)

Endpoint: POST /ingest/milestones

- Key fields:
 - employeeId
 - competencyId
 - targetLevel (L1/L2/L3)
 - dueDate
 - status (PENDING/ACHIEVED/WAIVED)
 - achievedDate (optional)
 - evidenceRef (optional)
 - sourceSystem
 - eventId

6.2 Employee Learning Profile Aggregation

A Learning Profile is an aggregated read-model that combines attendance statistics, assessment trends and milestone status per competency. It is recalculated on ingestion events and on scheduled batch evaluations to support near real-time monitoring.

6.3 Rule-based Risk Identification

Rules are defined externally (JSON/YAML) and versioned. Rules are evaluated at two scopes: EMPLOYEE and COMPETENCY. Each triggered rule produces an explainable risk event with severity, reason code and the data points used.

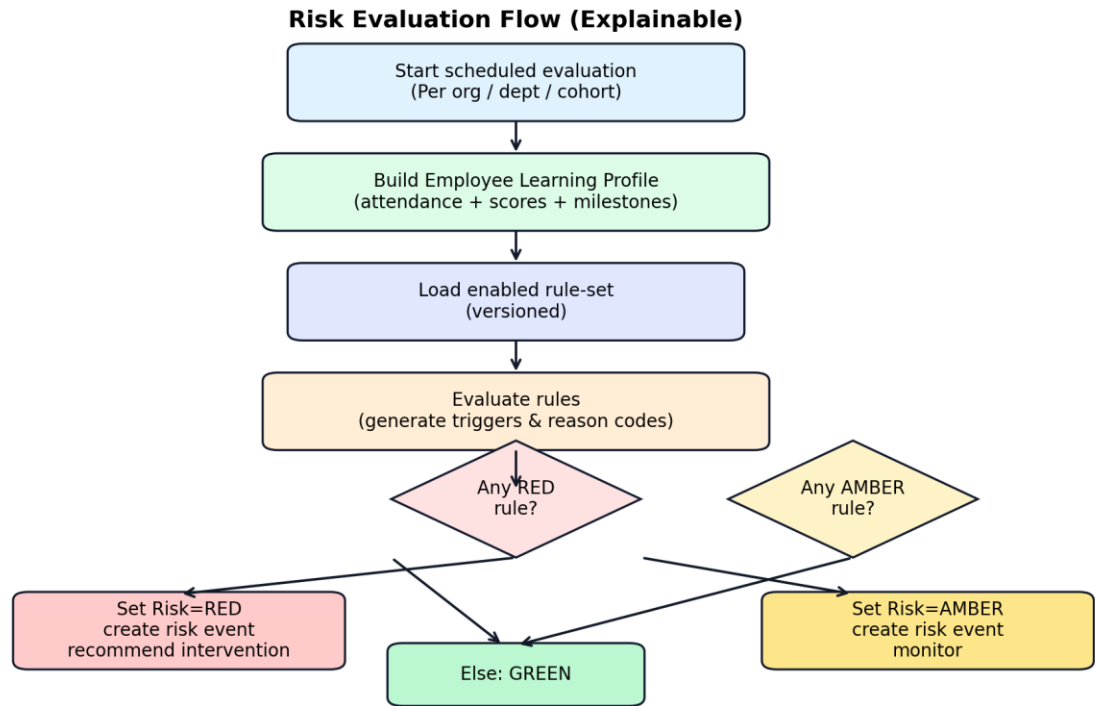


Figure 2: Explainable risk evaluation flow

6.3.1 Rule Definition Format (JSON)

```

{
  "ruleId": "R-ATTEND-001",
  "name": "Low attendance in last 30 days",
  "description": "Flag learner if attendance % for mandatory trainings is below threshold",
  "scope": "EMPLOYEE",
  "severity": "AMBER",
  "version": 1,
  "enabled": true,
  "conditions": {
    "windowDays": 30,
    "attendancePercent": { "lt": 75 },
    "trainingType": "MANDATORY"
  },
  "action": {
    "riskType": "ATTENDANCE_RISK",
    "reasonCode": "ATTENDANCE_BELOW_THRESHOLD",
    "recommendedInterventionTemplate": "REM-ATTEND-COACH"
  }
}

```

6.4 Intervention Tracking

When a learner is classified AMBER or RED, the system supports manual or auto-generated interventions. Interventions are tracked end-to-end, including owner, schedule, completion, outcome and follow-up evidence.

6.4.1 Intervention Types

- Remedial training session (targeted micro-learning / re-training)
- Coaching & mentoring assignment (mentor ownership, periodic check-ins)

6.4.2 Intervention Lifecycle

1. Created (manual/auto)
2. Assigned (trainer/mentor)
3. Scheduled
4. In-progress
5. Completed
6. Outcome recorded
7. Follow-up evaluation after N days

6.5 Dashboards and Compliance-ready Reporting

- Risk Dashboard: overall risk distribution, drill-down by department/location/role.
- Learner Profile View: timeline of attendance, assessments, milestones and interventions.
- Compliance Report: mandatory training completion with evidence and timestamps.
- Intervention Effectiveness: risk movement (RED→AMBER→GREEN), time-to-close and outcome rates.

7. Non-Functional Requirements

- Security: Role-based access control (RBAC) for L&D Admin, Trainer, Compliance, Employee.
- Auditability: Immutable audit log for data ingestion events, rule changes, intervention actions and report generations.
- Scalability: Support scheduled evaluations and near real-time updates; handle bursty ingestion.
- Reliability: Idempotent ingestion APIs (eventId), retries, dead-letter handling for failed events.
- Data Quality: Validation, deduplication and reconciliation dashboards for missing/late events.

8. Data Model (Conceptual)

Key entities and their relationships:

- Employee (1) → AttendanceRecord (many)

- Employee (1) → AssessmentResult (many)
- Employee (1) → CompetencyMilestone (many)
- Employee (1) → LearningProfile (1) [read model]
- LearningProfile (1) → RiskEvaluationResult / RiskEvent (many)
- RiskEvent (0..1) → Intervention (many) [one risk can have multiple interventions]
- Intervention (1) → InterventionOutcome (0..1)
- All actions → AuditLog (many)

9. Sample Realistic Risk Rules

9.1 Attendance rule

Attendance % for mandatory sessions in last 30 days < 75% → AMBER; < 60% → RED.

9.2 Assessment rule

Latest assessment score < 60% → AMBER; < 40% → RED; two consecutive failures triggers escalation.

9.3 Milestone overdue rule

Competency milestone overdue by 7+ days → RED.

9.4 Combined rule (escalation)

If milestone overdue AND assessment < 60% for same competency → RED with recommended remedial training + mentor assignment.

10. Demo Walkthrough (Suggested)

1. Open Risk Dashboard: show risk distribution (Green/Amber/Red) by department.
2. Select a RED learner: open consolidated learning profile with timeline view.
3. Open “Why flagged?”: show triggered rules, thresholds, and reason codes.
4. Assign intervention: choose remedial training or mentoring; set owner and due date.
5. Record completion and outcome: add notes and link evidence (new assessment).
6. Re-evaluate: show risk reduced (e.g., RED → AMBER/GREEN) and improved compliance.
7. Generate compliance report: export audit-ready report with timestamps and rule version used.

11. Future Enhancements

- ML-assisted risk prediction to complement rules while keeping rules as “explainable baseline”.
- Integration with HRIS for role-based required competencies and joiner/mover/leaver automation.
- Notification workflows (Teams/Email) for mentors and learners with escalation.
- Multi-tenant SaaS mode with tenant-level rule sets and data partitioning.
- Data retention policies aligned with corporate compliance requirements.